

Here we collect some basic facts relating independence of random variables with their characteristic functions. The main result is the following, whose proof requires two more theorems proved below.

Teorema 1. *Let $X = (X_1, \dots, X_n)$ and suppose that $\forall \xi \in \mathbb{R}^n$ $\phi_X(\xi) = \prod_{j=1, \dots, n} \phi_{X_j}(\xi_j)$. Then $X_1, \dots, X_n$ are independent.*

Proof. Let $Y = (Y_1, \dots, Y_n)$ vector of independent random variables, with $Y_j \sim X_j$ for $j = 1, \dots, n$. Now we prove $\phi_X = \phi_Y$, which will give us the thesis applying the unicity theorem.

$$\begin{aligned} \phi_Y(\xi) &= \mathbb{E}[\exp(i \cdot \sum_{j=1, \dots, n} \xi_j Y_j)] = \mathbb{E}[\prod_{j=1, \dots, n} \exp(i \xi_j Y_j)] \\ &= \prod_{j=1, \dots, n} \mathbb{E}[\exp(i \xi_j Y_j)] = \prod_{j=1, \dots, n} \phi_{Y_j}(\xi_j) = \prod_{j=1, \dots, n} \phi_{X_j}(\xi_j) \end{aligned}$$

By the unicity theorem we get ¹. $F_X = F_Y$, so in particular X_j 's must be independent since the Y_j 's are. □

Teorema 2. *Let $X, Y : (\Omega, \mathcal{A}) \rightarrow (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$. Suppose $\phi_X = \phi_Y$. Then $F_X = F_Y$.*

Proof. $\phi_X = \phi_Y$ means that $\forall \xi \in \mathbb{R}^d$ $\mathbb{E}[\exp(i \xi X)] = \mathbb{E}[\exp(i \xi Y)]$. Let $p(x) := \sum_{k=1, \dots, m} c_k \exp(i \xi_k x)$, where $c_k \in \mathbb{R}$ and $x, \xi_k \in \mathbb{R}^d$. Such an expression is called a trigonometric polynomial. We denote TP the set of all of them. By linearity of $\mathbb{E}$ we have $\mathbb{E}[p(X)] = \mathbb{E}[p(Y)]$. Consider $C_c(\mathbb{R}^d) := \{f : \mathbb{R}^d \rightarrow \mathbb{R} : f \text{ continuous with compact support}\}$. Observe that TP has the following properties (meaning that it satisfies Stone-Weierstrass theorem):

- TP algebra:
- TP separates: given $x \neq y \in \mathbb{R}^d$ one can find $f \in TP$ such that $f(x) \neq f(y)$.
- TP doesn't annihilate points: given $x \in \mathbb{R}^d$, consider any constant, non zero function, c . Then $c \in TP$ and $c(x) \neq 0$.
- TP closed under conjugate: $\bar{p}(x) = p(-x)$.

By Stone-Weierstrass theorem we have TP is dense in $C_c(\mathbb{R}^d)$ with the topology induced by the norm $\|\cdot\|_\infty$. Equivalently:

$$\forall g \in C_c(\mathbb{R}^d) \forall \epsilon > 0 \exists P_\epsilon \in TP \sup_{x \in \mathbb{R}^d} |g(x) - P_\epsilon(x)| < \epsilon$$

¹for any d -dimensional random variable X , define $F_X : \mathbb{R}^d \rightarrow [0, 1], (x_1, \dots, x_d) \mapsto \mathbb{P}(X_1 \leq x_1, \dots, X_d \leq x_d)$

Then:

$$\|\mathbb{E}[gX] - \mathbb{E}[gY]\| \leq \|\mathbb{E}[gX] - \mathbb{E}[P_\epsilon X]\| + \|\mathbb{E}[P_\epsilon X] - \mathbb{E}[P_\epsilon Y]\| + \|\mathbb{E}[gY] - \mathbb{E}[P_\epsilon Y]\| < \epsilon + 0 + \epsilon < 2\epsilon$$

Then $\|\mathbb{E}[gX] - \mathbb{E}[gY]\| < 2\epsilon$. Since ϵ is arbitrary we get $\forall g \in C_c(\mathbb{R}^d) \mathbb{E}[gX] = \mathbb{E}[gY]$. Apply now Riesz theorem to get μ_X, μ_Y probability measures on $\mathbb{R}^d$ of X, Y respectively. Then we have:

$$\int_{\mathbb{R}^d} g(x) \mu_X(dx) = \int_{\mathbb{R}^d} g(x) \mu_Y(dx)$$

and again by Riesz theorem we have $\mu_X = \mu_Y$, from which $F_X = F_Y$ follows. □

Teorema 3. *Let K be compact topological space. Let $\mathcal{A}$ subalgebra of $C(K, \mathbb{C})$. Suppose that $\mathcal{A}$ has the following properties:*

- *separates points: $\forall x \neq y \in K \exists f \in \mathcal{A} f(x) \neq f(y)$.*
- *never annihilates points: $\forall x \in K \exists f \in \mathcal{A} f(x) \neq 0$.*
- *closed under conjugate: $\forall f \in \mathcal{A} \bar{f} \in \mathcal{A}$.*

Then $\mathcal{A}$ dense in $C(K, \mathbb{C})$ with respect to uniform norm.

Proof. Consider $(P_n)_{n < \omega}$ defined by: $P_0 = c_0, P_{n+1}(t) = P_n(t) + \frac{t^2 - P_n(t)^2}{2}$. By induction on n one can prove that, for all $t \in [-1, 1]$, $P_n(t)$ is monotone increasing, bounded above by $t \mapsto |t|$. Then we can apply Monotone Convergence to get the punctual limit for $(P_n)_{n < \omega}$. Call it L . It follows that $L(t) = L(t) + \frac{t^2 - L(t)^2}{2}$. Then $L(t) = t^2$, from which we deduce (since $L \geq 0$, being $P_n \geq 0$ for all $n < \omega$) that $L(t) = |t|$. By Dini theorem, convergence is uniform. This means that we just found that $t \mapsto |t|$ can be approximated uniformly by polynomials. Now let $\bar{\mathcal{A}}$ be the closure of $\mathcal{A}$ with respect to $\|\cdot\|_\infty$. We now want to prove that $\bar{\mathcal{A}}$ is a lattice, or equivalently that $\max(f, g), \min(f, g) \in \bar{\mathcal{A}}$ for all $f, g \in \bar{\mathcal{A}}$. Observe that: $\max(f, g) = \frac{f+g+|f-g|}{2}$ and $\min(f, g) = \frac{f+g-|f-g|}{2}$. Since $\bar{\mathcal{A}}$ is a vector space, we have $f + g, f - g \in \bar{\mathcal{A}}$. We need to prove that if $h \in \bar{\mathcal{A}}$ then $|h| \in \bar{\mathcal{A}}$. Observe that, if $h \in C_c(\mathbb{R}^d)$ then h bounded. Let M be such that $\forall x \in K |h(x)| \leq M$. Let $p(t) = \sum_k c_k t^k$ be the polynomial that approximates uniformly $|t|$ in $[-M, M]$. Since $\bar{\mathcal{A}}$ algebra and $h \in \bar{\mathcal{A}}$ we get $\forall k < \omega h^k \in \bar{\mathcal{A}}$. Consequently $ph \in \bar{\mathcal{A}}$. Since ph uniformly approximates $|h|$, we have $|h| \in \bar{\mathcal{A}}$. In conclusion, $\bar{\mathcal{A}}$ is a lattice. Now let $f \in C(K), \epsilon > 0$. We want to find $g \in \bar{\mathcal{A}}$ such that $\|f - g\| < \epsilon$. First consider, for all $x \neq y \in K$, $h_{x,y} \in \mathcal{A}$ such that $h_{x,y}(x) = f(x), h_{x,y}(y) = f(y)$. Now fix y and for each x consider $h_{x,y}$. Since $h_{x,y}(x) = f(x)$ and $h_{x,y}, f$ continuous, we have that there exists U_x open neighbourhood of x such that

$\forall t \in U_x \ h_{x,y}(t) > f(t) - \epsilon$. Since $\{U_x\}_{x \in K}$ is an open covering of K and K is compact, we can extract a finite subcovering. Let $K \subseteq U_{x_1} \cup \dots \cup U_{x_m}$. Consider $g_y(t) := \max(h_{x_1,y}(t), \dots, h_{x_m,y}(t))$. Since $\overline{\mathcal{A}}$ lattice, we have $g_y \in \overline{\mathcal{A}}$. Furthermore the following hold:

- $\forall t \in K \ g_y(t) > f(t) - \epsilon$
- $g_y(y) = f(y)$

Since $g_y(y) = f(y)$ we can find (why?? g_y not continuous) V_y open neighbourhood of y such that $\forall t \in V_y \ g_y(t) < f(t) + \epsilon$. Again, as $\{V_y\}_{y \in K}$ is an open covering of K , compact, we can extract a finite subcovering $V_{y_1}, \dots, V_{y_k}$. Define $G(t) := \min(g_{y_1}(t), \dots, g_{y_k}(t))$. Since $\overline{\mathcal{A}}$ lattice, $G \in \overline{\mathcal{A}}$. Now we have that:

- $\forall i \ g_{y_i}(t) > f(t) - \epsilon$, so in particular $G(t) > f(t) - \epsilon$.
- $\forall t \in K \ \exists y_i \ t \in V_{y_i}$. Since $g_{y_i}(t) < f(t) + \epsilon$, we have $G(t) < f(t) + \epsilon$.

This means that $\forall t \in K \ f(t) - \epsilon < G(t) < f(t) + \epsilon$. Thus, $\|f - G\|_\infty < \epsilon$. Being ϵ arbitrary and $G \in \overline{\mathcal{A}}$, we have $f \in \overline{\mathcal{A}}$. Since f is an arbitrary element of $C(K)$, we conclude that $\overline{\mathcal{A}} = C(K)$.

□